Module 2

Chi-square tests to analyze categorical data

Topics

- Chi-square test for one categorical (or qualitative) variable.
- Chi square test for two categorical (or qualitative) variables.

Qualitative or Categorical Variables

Variables whose values represent labels or categories, rather than numerical quantities.

Examples:

Eye color: Black, Brown, Blue, and Green.
Nationality: US and Not US.
US Region: South, West, East, and North.

Chi-square test for one qualitative variable (also known as ***the chi-square goodness of fit test***)

- We have one categorical variable with K categories.
- We want to make an inference about the probability (percentage) associated with each category.
- We have an hypothesis about the possible values of these probabilities.
- We want to find out if the data support our hypothesized probabilities.

One-way table: Goodness of fit test

Example: Suppose three political candidates are running for the same elective position. Our interest is to validate the hypothesis that the three candidates are equally preferred. A sample of 150 voters is selected and their preference is recorded.

CANDIDATE		
1	**2**	**3**
61 Votes	53 Votes	36 Votes

One-way table: Goodness of fit test

1- Hypotheses:

Ho: $p_1 = c_1, p_2 = c_2, \ldots, p_k = c_k$

Where, $c_1 + c_2 + \ldots + c_k = 1$

Ha: At least one p_i is different from what we expect

Hypotheses (Example):

Ho: $p_1 = 1/3, p_2 = 1/3, p_3 = 1/3 \quad (1/3 + 1/3 + 1/3 = 1)$

Ha: At least one p_i is different from what we expect

One-way table: Goodness of fit test

2- Test statistic:

✓ E_i: The expected count for category i

$E_i = n\ p_i$

n: Sample size

p_i: Probability for category i

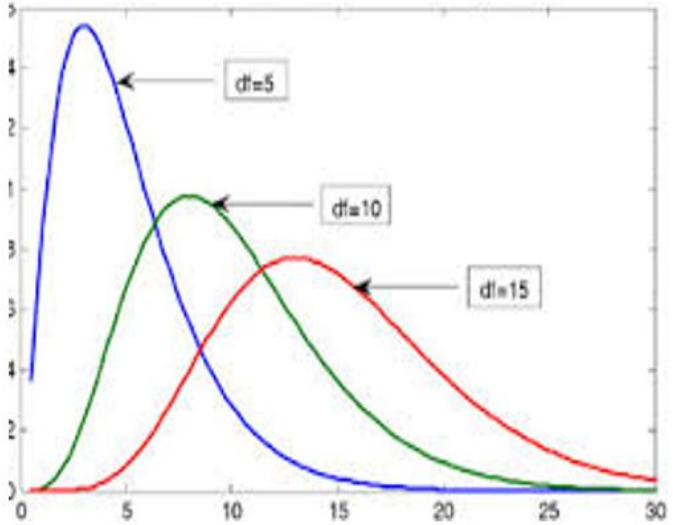
✓ O_i:The observed count for category i (number of observations in the sample for category i)

$$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}$$

One-way table: Goodness of fit test

2- Test statistic: χ^2 (Chi-square)

The χ^2 distribution



One-way table: Goodness of fit test

2- Test statistic (Example)

$p_1 = p_2 = p_3 = 1/3$

n= 150

$E_1 = E_2 = E_3 = 150 * 1/3 = 50$

	Candidate			
	1	2	3	
Oi	61	53	36	
Ei	50	50	50	**Total**
$\frac{(O_i - E_i)^2}{E_i}$	2.42	0.18	3.93	6.53

← $\chi^2 = 6.53$

$$\text{P value} = P(\chi^2 > 6.53)$$

Conditions required for a valid χ^2 one -way test

1-Take a random sample from the population of interest and measure the qualitative random variable for each element of the sample.

2-The sample size **n** will be large enough so that, for every category, the expected count E_i will be 5 or more.

Pearson residuals

When Ho is rejected and Ha is supported, the analyst usually wants to know *what categories are farther away from the expected probabilities and in what direction*. To do this, you can compute the *Pearson residuals.*

Pearson residuals measure how far each observed count is from its expected count.

Chi-square test for two qualitative variables (*chi-square test of independence*)

- We have two categorical variables and we want to know if they are related or not (i.e., dependent or independent?)
- Each variable has K categories.
- Each observation is classified according to the two variables. We create a **contingency table**. A table where the rows represent the levels (categories) of one variable and the columns represent the levels of the other variable.
- We compare the observed counts **(Oij)** for each combination of both variables to the count expected **(Eij)** when no association exists.

Example of chi-square test for independence

Researchers investigated the relationship between the gender of a viewer and the viewer's brand awareness. 300 TV viewers were randomly selected and each asked to identify products advertised by male celebrity spokespersons.

		Gender		
Brand awareness		Male	Female	**Total**
	Could identify product	95	41	136
	Could not identify product	50	114	164
	Total	145	155	300

Chi-square test for independence

1- Hypotheses:

Ho: The two variables are independent.

Ha: The two variables are dependent.

Hypotheses (Example):

Ho: The gender and the brand awareness of a person are independent.

Ha: The gender and the brand awareness of a person are dependent (the brand awareness of a person depends on the person's gender).

Chi-square test for independence

2- Test statistic: χ^2 (Chi-square) **DF = (# rows -1) * (# cols -1)**

✓ E_{ij}: The expected count for row i and column j

$$E_{ij} = \frac{R_i \, C_j}{n}$$

n: Sample size
R_i: Total for row i
C_j: Total for column j

✓ O_{ij} :The observed count for row i and column j (number of observations for row i and column j)

$$\chi^2 = \sum_{i=1}^{r} \sum_{j=1}^{c} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

Chi-square test for independence

2- Test statistic (Example)

		Gender		
		Male	Female	**Total**
Brand awareness	Could identify product	95	41	136
	Could not identify product	50	114	164
	Total	145	155	300

$$\hat{E}_{11} = \frac{R_1 C_1}{n} = \frac{(136)(145)}{300} = 65.73$$

$$\hat{E}_{12} = \frac{R_1 C_2}{n} = \frac{(136)(155)}{300} = 70.27$$

$$\hat{E}_{21} = \frac{R_2 C_1}{n} = \frac{(164)(145)}{300} = 79.27$$

$$\hat{E}_{22} = \frac{R_2 C_2}{n} = \frac{(164)(155)}{300} = 84.73$$

Chi-square test for independence

2- Test statistic (Example)

		Gender		
		Male	Female	**Total**
Brand awareness	Could identify product	95	41	136
	Could not identify product	50	114	164
	Total	145	155	300

$$\chi^2 = \frac{(95 - 65.73)^2}{65.73} + \frac{(41 - 70.27)^2}{70.27} + \frac{(50 - 79.27)^2}{79.27} + \frac{(114 - 84.73)^2}{84.73} = 46.14$$

Two-way table: Test for independence

When we **find** a **dependency** between the two variables, how do we **detect** the **categories** with higher **association**?

Similarly to the chi-square test for one variable, here you can also compute Pearson residuals.

We ***usually look for the cells with the largest residuals in absolute value*** because these cells correspond to categories with strong dependency (which could be a negative or positive dependency depending on the sign).

Chi-square test for two qualitative variables: **Practical significance.**

Cramer's V

- Cramér's V is a number between 0 and 1 that indicates how strongly two categorical variables are associated.
- Whereas a p-value close to zero means that the two variables have a statistically significant relationship (i.e., our variables are very unlikely to be completely unassociated), this does not mean the variables are strongly associated. For example, a weak association in a large sample size may also result in a p value very close to 0.

Chi-square test for two qualitative variables: **Practical significance.**

Cramer's V

It varies from 0 (= no association between the variables) to 1 (= complete association) and can reach 1 only when each variable is completely determined by the other.

Naive and simplified criteria to interpret Cramer's V:

C's V < 0.25, very weak association
0.25 < = C's V < 0.5 weak to moderate association
0.5 < = C's V < 0.75 moderate to strong association
C's V >= 0.75 strong association

$$\text{Cramer's V} = \sqrt{\frac{\chi^2}{n \cdot \min(r-1, c-1)}}$$

Validity assumptions for the chi-square test for independence

1-Take a random sample from the population of interest and measure the two qualitative random variables for each element of the sample.

2-The sample size **n** will be large enough so that, for every cell on the table, the expected count E_{ij} will be 5 or more.

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